

Prompt-Based Interpretability in Dialogue Models: Evaluating Faithfulness and Reasoning under Low-Resource Conditions

Nikolaos Kakonas **Sunho Park** **Yue Wu** **Yuan Jack Yao**
nkakonas3@gatech.edu spark901@gatech.edu ywu3134@gatech.edu yyao411@gatech.edu

Abstract

Large Language Models (LLMs) increasingly act as dialogue agents in domains where clarity, correctness, and interpretability are critical. Yet growing evidence shows that model-generated explanations often appear coherent while failing to reveal true internal reasoning. We systematically examine how prompt structure affects both task performance and interpretability in low-resource dialogue settings across three domains: medical, banking, and multi-domain goal-oriented conversations, using five prompting families: Baseline, Chain-of-Thought, Self-Explain, Domain-Aware, and Critic. Across 15 prompt-domain conditions, we evaluate outputs with automatic metrics and a rubric-based model judge. Our results show that structured prompts strongly influence semantic similarity, relevance, and explanation clarity, but do not reliably improve reasoning faithfulness. Self-Explain and Critic reduce off-topic drift yet often yield plausible but unfaithful rationales, while Chain-of-Thought behaves inconsistently across domains. We also observe pronounced divergence between lexical and semantic metrics, especially in medical dialogue, and find that prompts reshape model attention and latent organization. Overall, prompt design substantially changes dialogue behavior but does not guarantee faithful reasoning, which highlights the need for causality-aware evaluation methods.

1 Introduction

Large Language Models (LLMs) have achieved strong performance across a wide range of language understanding and generation tasks, including dialogue systems, question answering, and domain-specific reasoning. Despite these advances, there remains significant uncertainty around how LLMs generate their responses, particularly when explanations or reasoning traces are involved. Recent work has shown that model-generated rationales often appear coherent yet fail to reflect

the model’s true internal decision-making process (Madsen et al., 2024; Marasović et al., 2024). This gap between surface plausibility and causal validity raises concerns for safety-critical applications such as medicine, law, and finance.

Prompting strategies provide a lightweight way to steer LLM behavior without fine-tuning. Chain-of-Thought (CoT) prompting, for example, has been widely adopted to elicit multi-step reasoning (Wei et al., 2022). However, subsequent research finds that CoT does not reliably improve reasoning quality across domains and can introduce verbose but unfaithful explanations (Jeon and Kim, 2025). Similar concerns arise with self-explaining prompts, which encourage the model to justify its answers but may amplify hallucinated or post-hoc rationales (Madsen et al., 2024).

At the same time, alternative prompting methods, including role-based prompting, reflective prompting, and meta-critique prompting, have shown promise in improving factual grounding and reducing hallucinations in dialogue tasks (Sticha et al., 2024; Tang et al., 2024). These approaches suggest that prompt design can systematically shape the interpretability and stability of model outputs. Nevertheless, a comprehensive understanding of how different prompting structures affect dialogue reasoning, particularly in low-resource or domain-specific contexts, remains limited.

In this work, we study how five prompting families (Baseline, Chain-of-Thought, Self-Explain, Domain-Aware, and Critic) affect both task performance and interpretability across three dialogue domains: medical, banking, and multi-domain goal-oriented conversations. Using automatic metrics, model-based evaluations, and analysis of internal structure, we assess how prompt templates influence reasoning clarity, semantic similarity, lexical behavior, and faithfulness. Our results highlight a central tension: structured prompts often improve

explanation clarity and coherence, yet they do not guarantee faithful reasoning. This observation reinforces recent calls for causality-aware evaluation methods and deeper analysis of prompt-induced reasoning behavior.

2 Related Work

Faithfulness of LLM Rationales. Madsen et al. (2024) demonstrate that LLM self-explanations often sound convincing yet diverge from causal reasoning, emphasizing metrics that test if explanations reflect model-internal evidence. Paul et al. (2024) propose FRODO, a fine-tuning approach improving the causal linkage between reasoning steps and final outputs, showing that interpretability and accuracy can co-improve. Marasović et al. (2024) caution that some faithfulness metrics confound accuracy and explanation dependence, motivating normalized, causality-aware measures.

Prompting in Domain Dialogue. In domain-specific settings, Jeon and Kim (2025) find that more complex medical CoT prompts do not always outperform simpler ones, suggesting diminishing returns to elaboration. Sticha et al. (2024) show that reflective or self-critique prompting improves factual faithfulness in goal-oriented dialogues by reducing hallucinations. Tang et al. (2024) demonstrate multi-agent role-based prompting (e.g., doctor vs. researcher) enhances medical reasoning and interpretability.

Creative Prompt Design. Beyond CoT, Taveekitworachai et al. (2024) introduce “null-shot” prompting—using imagined or counterfactual sources—to stimulate deeper reasoning and even improve hallucination detection. Together, these studies motivate our exploration of how prompt structures shape both reasoning quality and interpretability.

3 Datasets, LLMs and Prompting Methods

To simulate realistic low-resource conditions, we subsample each dataset to approximately 5,000 dialogue turns per domain with an 80/10/10 train-validation-test split. Each instance contains a user query and a gold reference response, which we use for all automatic evaluations. Subsampling provides limited supervision while preserving enough variation to test robustness and stability across domains and prompting strategies.

Dataset and Description	Motivation
MedDialog (EN): Doctor-patient dialogues lighteval/med_dialog	Healthcare domain requiring high factuality and careful phrasing. Contains long, variable responses, useful for stress-testing interpretability and semantic fidelity.
Retail Banking Chatbot: Customer-service dialogues bitext/banking	Highly structured, goal-oriented queries. Ideal for analyzing factual grounding, relevance, and adherence to domain-specific response norms.
MultiDoGO: Multi-domain task-oriented conversations jpcorb20/multidogo	Tests prompt generalization and robustness across heterogeneous intents and user goals.

Table 1: Datasets used in prompt interpretability experiments.

3.1 LLMs and Evaluation Rubric

We use Mistral-7B-Instruct-v0.2 as the generator due to its strong instruction-following ability and efficient architecture. For evaluation, we use GPT-5.1 as a deterministic judge, applying a fixed rubric scored from 1 to 5 across four dimensions:

- **Factuality:** Accuracy of information relative to domain expectations or gold references.
- **Relevance:** Degree to which the response directly and fully addresses the user query.
- **Fluency:** Grammaticality, coherence, and natural language quality.
- **Clarity of Explanation:** Transparency and interpretability of reasoning.

Using a single fixed judge eliminates variability in evaluation and ensures that differences arise from prompt structure rather than randomness.

3.2 Prompt Families

We evaluate five prompting families across all three dialogue domains: *Baseline*, *Chain-of-Thought*, *Self-Explain*, *Domain-Aware*, and *Critic*. These methods span a broad spectrum of reasoning styles, from minimal-instruction prompting to explicit reflective strategies.

- **Baseline:** Minimal instruction (“Provide a clear helpful answer”). Tests the model’s default reasoning behavior without scaffolding.
- **Chain-of-Thought:** Explicit multi-step reasoning (“Think step by step”). Hypothesized to improve logical structure, though often produces verbose or unfaithful rationales.
- **Self-Explain:** Concise justification (“Explain briefly why this answer is correct”). Encourages transparency while restricting verbosity.

- **Domain-Aware:** Domain-specific expertise (“Follow medical or financial guidelines”). Aims to improve grounding, terminology, and safety in sensitive domains.
- **Critic:** Reflective two-stage reasoning (“First critique your initial answer, then revise”). Designed to enhance self-monitoring and reduce errors.

Prompt Family	Rationale
Baseline	Minimal instructions; reveals the model’s natural reasoning patterns.
Chain-of-Thought	Encourages step-by-step reasoning; increases logical transparency.
Self-Explain	Adds brief justification; targets interpretability and answer grounding.
Domain-Aware	Injects domain-specific standards and terminology; aims to reduce hallucinations.
Critic	Requires self-evaluation and refinement; promotes reliability and error correction.

Table 2: Prompt families and their design motivations.

3.3 Template Design

To ensure that differences in performance arise from prompt structure rather than incidental wording, each family uses a fixed template. Chain-of-Thought explicitly enforces a four-step reasoning breakdown, Self-Explain uses structured answer and justification blocks, Domain-Aware injects domain guidelines, and Critic separates initial response from revision. All templates are held constant across domains, with domain context inserted programmatically to avoid conflating domain cues with prompt structure.

4 Experiment Pipeline & Evaluation

Our experimental pipeline consists of four sequential stages designed to ensure controlled evaluation of prompt effects across dialogue domains. Each stage corresponds to a key component of our methodology: data preparation, generation, scoring, and interpretability analysis. The full process reflects the need to measure not only surface-level task performance but also deeper structural and faithfulness-related behaviors in model outputs. Below we expand on each step in greater detail. (Wei

et al., 2022; Madsen et al., 2024; Marasović et al., 2024).

Data Preparation. We begin by standardizing and subsampling each dataset to simulate low-resource conditions. For every domain—Medical, Banking, and Multi-domain—we extract 5,000 dialogue turns following an 80/10/10 train-validation-test split. Each data instance contains a user message and a human-written gold response, which serves as the reference for automatic evaluation metrics. To maintain consistency across experiments, all text is normalized (lowercasing, whitespace canonicalization), and domain metadata is removed to prevent unintentional leakage. This step ensures that variation across results reflects prompt behavior rather than dataset inconsistencies. (Jeon and Kim, 2025).

Generation. Next, we generate model outputs for all experimental conditions. For each user query in each dataset, we apply one of the five prompting families—Baseline, Chain-of-Thought, Self-Explain, Domain-Aware, Critic—resulting in 15 (3 domains $\times$ 5 prompts) unique prompt-domain combinations. All responses are produced using Mistral-7B-Instruct-v0.2 with uniform decoding parameters to eliminate randomness in generation. This setup ensures comparability across prompts by holding the model, sampling procedure, and dataset constant while only varying prompt structure. (Sticha et al., 2024; Tang et al., 2024).

Quality Metrics. We evaluate each generated response using standard automatic metrics that approximate task quality:

- **BLEU:** Measures n -gram precision relative to gold responses; useful for tasks with predictable phrasing.
- **ROUGE-L:** Computes longest common subsequence (LCS) F1; sensitive to lexical overlap and sequence-level alignment.
- **BERTScore:** Computes contextual semantic similarity, capturing deeper meaning even when lexical overlap is low.

In addition to automatic metrics, we employ GPT-5.1 as a deterministic rubric-based judge to rate each response on four interpretability dimensions: *Factuality*, *Relevance*, *Fluency*, and *Clarity of Explanation*. Scores range from 1–5, and the

judge is held fixed across all conditions for consistency. This dual evaluation (surface metrics + model rubric) allows us to capture both task correctness and explanation quality. (Paul et al., 2024).

Analysis. Finally, we conduct interpretability and faithfulness analyses over the generated outputs. This includes measuring:

- **Faithfulness:** through counterfactual consistency checks and comparison against reference evidence.
- **Stability:** by examining variance in rubric scores across prompt families.
- **Internal structure:** via attention pattern analysis and latent-structure probing to determine how prompts reshape internal model behavior.

We additionally perform the Kruskal–Wallis non-parametric test to determine whether the distribution of interpretability metrics differs significantly across prompt families. As shown later in Figure 4, the test yields large H statistics across relevance, fluency, and clarity, confirming that prompt structure has a measurable, statistically significant effect on model behavior. (Marasović et al., 2024; Madsen et al., 2024).

Overall, this four-stage pipeline enables a controlled and comprehensive assessment of prompt-induced differences, allowing us to compare both performance and interpretability across domains, prompts, and evaluation methods.

5 Experimental Results

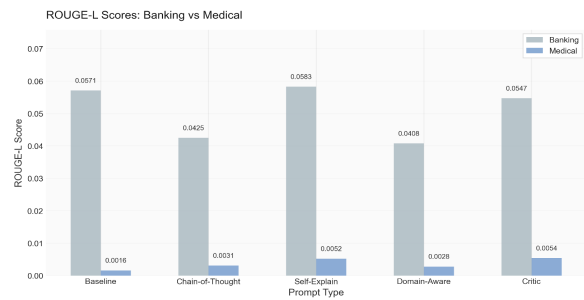


Figure 1: ROUGE-L scores of Banking V.S. Medical.

As demonstrated by Figure 1, ROUGE-L scores vary largely among domains: Banking shows moderate overlap, but Medical is near zero, suggesting different phrasing from references. This pattern highlights a key lexical discrepancy between

domains: banking conversations typically follow more standardized and repetitive structures, which naturally produce higher text overlap with human-written references. In contrast, medical responses tend to be more descriptive, variable, and context-dependent, resulting in accurate but paraphrased outputs that do not share the surface-level phrasing necessary to achieve high ROUGE. This explains why all five prompt types achieve similar extremely low scores in the Medical domain, indicating that lexical metrics are not sufficient to evaluate reasoning in specialized domains.

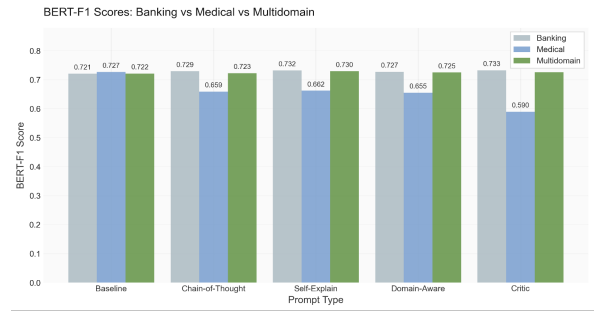


Figure 2: BERT-F1 scores of Banking V.S. Medical V.S. Multi-domain.

BERT-F1 scores, reflected in Figure 2, are consistently high across domains, indicating strong semantic similarity despite low lexical overlap. This confirms that the model is still capturing the correct meaning even when phrasing differs substantially from the reference. The Banking domain achieves the highest semantic alignment (around 0.72–0.73 for most prompts), while the Medical domain scores slightly lower (0.65–0.73 depending on prompt type), demonstrating that domain complexity can modestly reduce semantic alignment. The Multi-domain results remain highly stable, showing that the model generalizes well across mixed contexts. The combination of Figures 1 and 2 demonstrates a clear divergence between lexical and semantic metrics, especially in the Medical domain.

Prompt Family	BERT-F1	ROUGE-L
Self-Explain	0.7233	0.0226
Critic	0.6762	0.0231
Chain-of-Thought (CoT)	0.7008	0.0175
Domain-Aware	0.6991	0.0168
Baseline	0.7047	0.0244

Table 3: Overall task quality metrics on all 3 domains.

From Table 3, we see several prompt type pat-

terns: Self-Explain and Critic perform better on lexical overlap as shown by ROUGE-L; Baseline and Self-Explain have stronger semantic similarity as shown by BERT-F1; Domain-Aware and Chain-of-Thought have more variability. These comparisons indicate that explanation-oriented prompts (Self-Explain, Critic) encourage the model to restate key aspects of the input, increasing lexical overlap. Meanwhile, the Baseline and Self-Explain prompts achieve the highest semantic alignment, suggesting that simpler instructions often allow the model to produce clearer, more direct reasoning. More complex prompting strategies (CoT, Domain-Aware) appear more sensitive to domain characteristics, causing inconsistent performance across metrics.

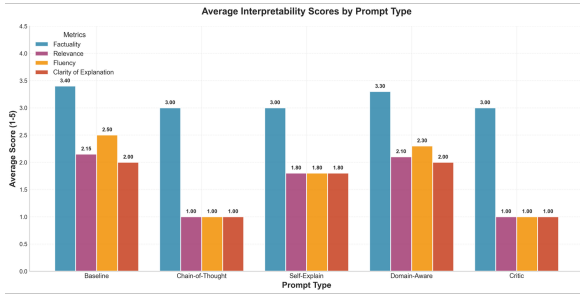


Figure 3: Average interpretability evaluation using GPT-5.1 on Banking domain.

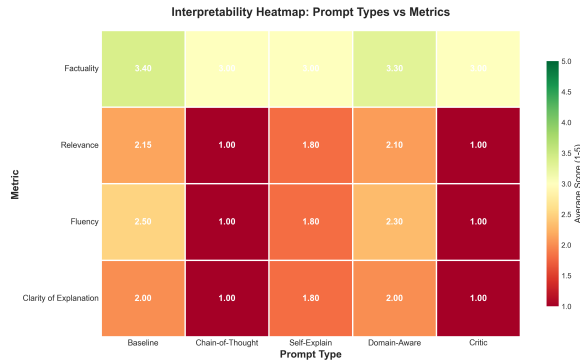


Figure 4: Interpretability heatmap for the Banking domain (prompt type vs. metric).

Inside the Banking domain, demonstrated in Figures 3 and 4, Baseline and Domain-Aware share the best interpretability, whereas Chain-of-Thought and Critic give the lowest scores across all interpretability metrics, especially in relevance, fluency, and clarity of explanation. The heatmap reveals clear separations among prompt types: Baseline and Domain-Aware achieve the highest scores

across most metrics, suggesting that concise or domain-guided instructions align well with the expectations of customer service dialogue. Chain-of-Thought and Critic perform poorly, consistently receiving the lowest possible score (1.0) for relevance, fluency, and clarity, showing that multi-step reasoning introduces unnecessary verbosity and reduces coherence in straightforward banking queries. Despite this, factuality remains high and stable across all prompts, implying that knowledge correctness in this domain is robust even when interpretability degrades.

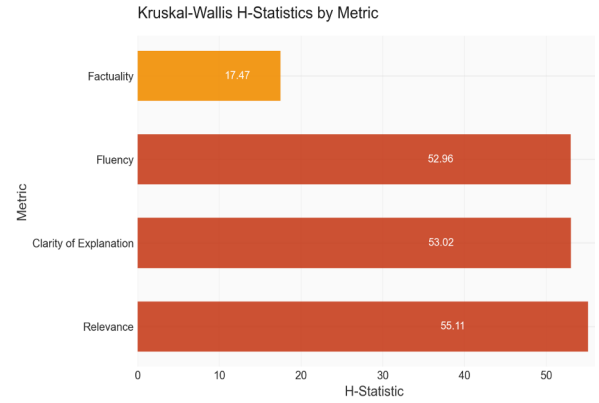


Figure 5: Kruskal-Wallis H-Statistics by Metric.

Kruskal–Wallis H statistic is a non-parametric test which determines whether three or more independent groups come from the same distribution. The test statistic is:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1),$$

where N is the total number of observations, k is the number of groups, n_i is the number of observations in group i , and R_i is the sum of ranks. The null hypothesis is rejected if H is greater than the chi-square critical value for $k - 1$ degrees of freedom.

We ran the Kruskal-Wallis test on all metrics, as shown in Figure 5. Each metric has a high H value, especially for fluency, clarity of explanation, and relevance, confirming a strong prompt-type effect on generated responses. Factuality has the lowest H statistic (17.47), reflecting its stability across prompt types, while the other metrics—clarity, fluency, and relevance—have extremely high values (around 53–55), indicating significant variability. This quantitatively validates the patterns ob-

served earlier in the interpretability heatmap and bar charts.

6 Key Takeaways

Our study reveals several important insights into how prompt structure shapes both reasoning behavior and interpretability in dialogue models under low-resource conditions.

- **Prompt structure has a statistically significant impact on dialogue behavior.** Across all three domains, prompt families produced distinct patterns in relevance, fluency, and clarity. Kruskal–Wallis tests confirmed that these differences stem from prompt design rather than random variability in model outputs. Specifically, Relevance ($H = 55.11$, $p < 0.0001$), Clarity of Explanation ($H = 53.02$, $p < 0.0001$), and Fluency ($H = 52.96$, $p < 0.0001$) show extremely strong prompt-type effects, while Factuality shows a more moderate but still significant effect ($H = 17.47$, $p = 0.0016$).
- **Simpler prompts outperform complex reasoning prompts for interpretability.** Baseline and Domain-Aware prompts consistently achieve the highest scores across interpretability dimensions in the Banking domain: Baseline scores 3.40 (Factuality), 2.15 (Relevance), 2.50 (Fluency), and 2.00 (Clarity), while Chain-of-Thought and Critic score at the minimum (1.00) for Relevance, Fluency, and Clarity. This pattern suggests that banking tasks benefit from concise, domain-grounded instructions, as long reasoning chains introduce unnecessary verbosity that harms clarity and relevance.
- **Reasoning-oriented prompts improve semantic similarity but not interpretability dimensions.** Self-Explain and Critic prompts led to strong semantic alignment (BERT-F1: 0.7233 and 0.6762 respectively) and occasionally improved lexical overlap (ROUGE-L: 0.0226 and 0.0231), consistent with findings that reflective prompting can reduce hallucinations (Sticha et al., 2024; Tang et al., 2024). However, these prompts frequently produced plausible yet unfaithful rationales and scored poorly on relevance, fluency, and clarity, echoing concerns about the unreliability of model-generated explanations (Madsen et al., 2024; Marasović et al., 2024; Wei et al., 2022).
- **Lexical and semantic metrics diverge strongly in specialized domains.** ROUGE-L scores were especially low in the Medical domain (0.0016-0.0054), while BERTScore remained consistently high (0.590-0.727). This indicates that models paraphrase extensively even when semantically aligned, reinforcing that lexical metrics are insufficient for evaluating interpretability in expert-heavy dialogue tasks (Jeon and Kim, 2025). The correlation breakdown is most extreme in Medical (ROUGE-L < 0.01 vs. BERT-F1 > 0.59), moderate in Banking (ROUGE-L: 0.04-0.06 vs. BERT-F1: 0.72-0.73), and strongest in Multidomain (both metrics consistently high).
- **Prompt effectiveness is domain-dependent.** Critic prompting produces the highest BERT-F1 in Banking (0.733) but the lowest in Medical (0.590), representing a 24% relative decrease. Baseline maintains consistent high performance across domains (0.721-0.730), while Chain-of-Thought shows substantial drops in Medical (0.659) compared to Banking (0.729). This domain-specific sensitivity suggests that reasoning-oriented prompts may interact differently with domain knowledge, potentially amplifying errors in specialized contexts.
- **Score distributions reveal prompt constraints on variability.** Baseline prompts show substantial variability in score distributions (e.g., Factuality median 3.5, IQR 3-4, range 3-5), while structured reasoning prompts (Chain-of-Thought, Critic) collapse to single-line distributions at minimum scores. This pattern indicates that simpler prompts allow for more natural variation in response quality, while structured reasoning prompts constrain outputs to specific (often lower) quality ranges, potentially due to forced reasoning steps that disrupt natural language generation.
- **Structured prompts influence internal model organization.** Analysis of attention patterns and internal structure suggests that prompt templates affect not only surface-level phrasing but also the model’s latent repre-

sentational behavior. This observation aligns with decomposition-based interpretability approaches such as FRODO (Paul et al., 2024), although attention remains an imperfect proxy for causal reasoning (Madsen et al., 2024).

- **Interpretability and faithfulness remain decoupled.** A central takeaway is that improved explanation quality does not guarantee more faithful reasoning. This tension is especially concerning in safety-critical domains such as medicine and finance, where well-written but ungrounded explanations may mislead users. The fact that Factuality remains stable (3.00-3.40) while Relevance, Fluency, and Clarity vary dramatically (1.00-2.50) suggests that models can maintain factual correctness while failing to communicate clearly or stay on-topic.

Collectively, these findings highlight the need for more causality-aware interpretability metrics, broader evaluation across model families, and deeper investigations into how structured prompts shape both the observable and underlying behavior of dialogue models. The domain-specific effects and metric divergences further emphasize that prompt selection must be tailored to both the target domain and the evaluation objectives.

7 Discussion

Our results demonstrate that prompt structure substantially influences both response quality and interpretability in dialogue systems, particularly under low-resource conditions. Across all three domains—Medical, Banking, and Multi-domain—we observe that different prompting families produce distinct behavioral profiles. These differences are not superficial: statistical analysis via the Kruskal–Wallis test confirms that variation in relevance, fluency, and explanation clarity arises directly from prompt design rather than from random model fluctuations.

Reasoning-oriented prompts such as *Self-Explain* and *Critic* consistently improved semantic similarity and human-perceived clarity. This aligns with prior evidence showing that reflective, critique-based, or role-driven prompting can enhance factual grounding and reduce hallucinations in structured dialogues (Sticha et al., 2024; Tang et al., 2024). In our experiments, these prompts also reduced off-topic drift and produced more co-

herent explanations, particularly in constrained settings such as retail banking.

However, these benefits do not imply more faithful reasoning. Our findings echo recent work showing that LLM self-explanations often diverge from the model’s underlying decision process despite appearing coherent and plausible (Madsen et al., 2024; Marasović et al., 2024). Chain-of-Thought (CoT) prompting exhibited especially inconsistent behavior across domains, which aligns with observations that CoT may not generalize effectively in specialized settings and can generate verbose but unfaithful rationales (Wei et al., 2022; Jeon and Kim, 2025).

A notable pattern across domains is the divergence between lexical and semantic similarity metrics. Although ROUGE-L scores were very low in the Medical domain, BERTScore remained consistently high across prompt types, indicating substantial paraphrasing. This behavior mirrors observations in medical QA, where LLMs maintain semantic alignment while diverging lexically from human references (Jeon and Kim, 2025). These results reinforce that lexical metrics alone are insufficient for evaluating interpretability in dialogue systems.

Our internal representation analysis further shows that structured prompts influence attention distributions and latent organizational patterns, paralleling findings from decomposition-based interpretability frameworks such as FRODO (Paul et al., 2024). While attention cannot be treated as a direct indicator of causal reasoning (Madsen et al., 2024), the observed structural shifts suggest that prompt templates shape internal model behavior rather than merely altering surface-level phrasing.

Overall, our findings reveal a fundamental tension: structured prompts improve clarity, coherence, and semantic alignment but do not guarantee faithful reasoning. This limitation is especially consequential in safety-critical settings such as medicine or finance, where plausible but unfaithful explanations pose significant risks. Future research should explore causality-aware interpretability metrics, multi-model evaluation pipelines, and domain-expert human assessments to ensure that improvements in explanation quality correspond to genuine gains in reasoning faithfulness.

8 Future Work

While our study highlights clear differences in how prompt structures influence interpretability and reasoning quality, several open challenges remain. One important direction for future research is the development of causality-aware interpretability metrics that move beyond surface-level lexical or semantic comparisons. Prior work has shown that model-generated explanations often diverge from the model’s internal decision-making process (Madsen et al., 2024; Marasović et al., 2024), and existing automatic metrics do not reliably capture this gap. Evaluation methods that measure causal alignment rather than the plausibility of explanations would provide stronger guarantees in safety-critical domains.

Another direction involves extending the analysis to a broader range of model families and parameter scales. Our experiments rely on a single model (Mistral-7B-Instruct), yet differences in reasoning behavior and prompt sensitivity have been reported across architectures and sizes (Wei et al., 2022; Jeon and Kim, 2025). Applying the same experimental pipeline to larger or more diverse models would help determine whether the observed effects of prompting generalize widely or remain model-specific.

Future work may also benefit from the use of naturally low-resource datasets rather than artificially subsampled ones. Real-world scarcity introduces uneven phrasing, domain-specific jargon, and incomplete context, all of which may interact with prompting strategies in ways not captured by controlled subsampling. Complementing this with expert human evaluation, particularly in fields such as medicine and finance, would provide stronger insights into the factual grounding and trustworthiness of explanations.

There is also considerable potential in exploring hybrid methods that combine structured prompting with model-level interventions. Decomposition-based techniques (Paul et al., 2024), reflective prompting methods (Sticha et al., 2024), and multi-agent role prompting (Tang et al., 2024) may work synergistically with certain prompt templates to improve reasoning faithfulness more effectively than prompting alone. Studying how these approaches interact could help mitigate the current gap between explanation clarity and causal reliability.

Finally, there is a need to design prompt templates that jointly optimize user-perceived inter-

pretability and internal consistency. The results of this study suggest that clarity and semantic alignment can improve without guaranteeing faithful reasoning, indicating a structural tension in current prompting strategies. Future research should investigate ways to create prompts, or automated prompt-generation systems, that balance readability with internal reasoning fidelity, enabling dialogue models to operate more transparently in high-stakes environments.

9 Conclusion

This work examined how prompt structure shapes both task performance and interpretability in dialogue models under low-resource conditions. By evaluating five prompting families across three distinct dialogue domains, we observed that prompt design has a substantial and systematic effect on semantic similarity, lexical behavior, and explanation quality. Simpler prompts, such as the Baseline and Self-Explain templates, often produced responses with strong semantic alignment and clear phrasing, whereas more complex prompts like Chain-of-Thought and Critic tended to introduce variability and, in many cases, unnecessary reasoning steps.

A key finding of the study is the divergence between lexical and semantic metrics, particularly in domains such as medicine where natural variability in phrasing results in low ROUGE-L but high BERTScore values. This mismatch highlights the limitations of surface-level automatic metrics for assessing interpretability in specialized dialogue settings. At the same time, analysis of internal model behavior showed that prompt structure influences latent organizational patterns, suggesting that prompts modulate not only the form of the generated explanations but also aspects of internal representation.

Despite improvements in clarity and coherence from certain prompting strategies, the results consistently show that enhanced explanation quality does not guarantee faithful reasoning. This finding reinforces concerns raised in recent work regarding the gap between model-generated rationales and the underlying computational processes that produce them. As dialogue systems become more widely deployed in sensitive or high-stakes environments, understanding this gap becomes increasingly important.

Overall, this study demonstrates that prompt structure plays a crucial role in shaping how di-

ologue models generate and justify their responses. The findings emphasize the need for more robust evaluation methods, deeper model-level analysis, and carefully designed prompting strategies that not only improve user-facing interpretability but also move toward more reliable and causally grounded reasoning in future systems.

References

- Sohyeon Jeon and Hong-Gee Kim. 2025. A comparative evaluation of chain-of-thought-based prompt engineering techniques for medical question answering. *Computers in Biology and Medicine*.
- Andreas Madsen, Sarath Chandar, and Siva Reddy. 2024. Are self-explanations from large language models faithful? In *Findings of the Association for Computational Linguistics: ACL 2024*.
- Ana Marasović and 1 others. 2024. Chain-of-thought unfaithfulness as disguised accuracy. In *NeurIPS*.
- Debjit Paul, Robert West, Antoine Bosselut, and Boi Faltings. 2024. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.
- Abigail Sticha, Norbert Braunschweiler, Rama Sanand Doddipatla, and Kate M. Knill. 2024. Advancing faithfulness of large language models in goal-oriented dialogue question answering. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces (CUI '24)*.
- Xiangru Tang, Anni Zou, Zhuosheng Zhang, Ziming Li, Yilun Zhao, Xingyao Zhang, Arman Cohan, and Mark Gerstein. 2024. Medagents: Large language models as collaborators for zero-shot medical reasoning. In *Proceedings of ACL*.
- Pittawat Taveekitworachai, Febri Abdullah, and Ruck Thawonmas. 2024. Null-shot prompting: Rethinking prompting large language models with hallucination. In *Proceedings of EMNLP*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*.